Decentralized Agentic Edge Traffic Management System for Predictive Emergency Preemption

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Abstract—**Current intelligent traffic management systems (ITMS) mostly rely on centralized cloud computing, which introduces significant latency, single points of failure, and privacy concerns from transmitting and storing raw visual data. This paper introduces the Decentralized Agentic Edge (DAE) framework, a novel approach by leveraging edge computing hardware and on device artificial intelligence. DAE processes traffic camera feeds locally using a TensorRT optimized YOLO26 model for real time vehicle classification and density estimation. A hierarchical Multi Agent System (MAS) runs directly on edge hardware dynamically adjusts green light durations using local data. For emergency vehicles, it enables predictive preemption using lightweight, asynchronous, Infrastructure to Infrastructure (I2I) communication protocol over an offline MQTT mesh network. Experimental results using an NVIDIA Jetson Nano device demonstrate the system's viability, achieving a processing latency under 50 ms (average 42 ms) with low resource utilization, enabling rapid local decision making. Simulations show a drop in civilian wait times (24.5%) and, significant 93% reduction (from 34 s to 2.1 s) in emergency vehicle delay compared to reactive models. The DAE framework offers a resilient, low latency, privacy centric alternative to centralized ITMS, capable of transitioning urban traffic management from reactive control to predictive optimization.**

Keywords: ***Edge Computing, Traffic Management, Artificial Intelligence (AI), Computer Vision,MQTT , Wireless Mesh Networks, Predictive Control, Decentralized Systems, Intelligent Transportation Systems (ITS), Real Time Systems.***

I. INTRODUCTION

Traffic congestion is one of the major issues facing urban areas today resulting in time wastage, increased use of fuel, and lowered quality of life. The traditional system of traffic lights with predefined timings has been found to be inadequate in addressing these challenges. The use of Intelligent Traffic Management Systems (ITMS) will bring about effective management of traffic. [1], [2].

However, existing ITMS solutions often face critical limitations:

- **Latency Bottlenecks:** Cloud based centralized architecture where the cloud server transfers the sensory data (usually high definition videos) through network connections suffers from substantial round trip latencies (several hundred milliseconds to several seconds) before any action is taken [2], [6]. This latency is incompatible with the need for traffic signal control.
- **Single Point of Failure:** The reliance on cloud technology implies that the whole traffic infrastructure becomes susceptible to internet downtime, server crashes, or any cyber attacks. All of the above might lead to traffic jams and, thus, the necessity to return to less efficient static timers. [6].
- **Privacy Concerns:** Recording and processing the video stream from the cameras entails serious privacy concerns regarding surveillance and personally identifiable information (PII) [7]. While masking techniques exist, they often add latency or need centralized processing.
- **Reactive Emergency Handling:** Current adaptive systems either cloud based or human controlled respond reactively to emergencies. They can only trigger a green phase when an emergency vehicle is detected within the sensor range, which may leave the emergency responders trapped behind traffic that is moving slowly. [3], [4].
- **Scalability and Cost:** Centralized systems require high bandwidth network infrastructure and powerful backend servers, which in turn raises both deployment and operational expenses. [6].

This paper addresses these limitations by proposing the **Decentralized Agentic Edge (DAE)** framework. DAE utilizes edge computing devices strategically placed at intersections. Each device performs critical tasks locally:

- **On Device AI:** A TensorRT optimized YOLO26 model processes camera feeds in real time for vehicle detection, classification, and density estimation [8].
- **Edge Multi Agent System (MAS):** a light weight MAS is run directly on the edge hardware, using the results from the AI to control the timing of traffic signals in favor of both civilian and detected emergency vehicles.
- **Decentralized Communication:** An efficient and compact MQTT mesh network for I2I communication helps achieve predictive proactive coordination. In case an upstream intersection identifies an approaching emergency vehicle, it could announce its identification information and trajectory data to downstream intersections. [14].

The key contributions of this paper are:

- **Novel Architecture:** Proposing the DAE framework as a model shift from centralized ITMS to a fully decentralized, edge optimized system.

- **Performance Demonstration:** Experimentally validating the system's low latency processing capability (less than 50 ms) and efficiency on standard edge computing hardware (Jetson Nano) [10].
- **Predictive Capability:** Proving the viability of predicting and preempting emergencies through a lightweight I2I protocol [5].
- **Privacy by Design:** Creating intelligent traffic infrastructure where data processing is masked by default, eliminating the need for storage or transfer of raw video information [7].

II. Related Work and Comparative Analysis

There have been advancements made in the field of intelligent traffic management, with technology having advanced from merely mechanical timers to more digital systems. The existing techniques for traffic management may broadly be categorized into:

- **Static Timer Based Systems:** This is an outdated approach based on pre planned cycles. They do not possess adaptive capabilities to changing traffic conditions and suffer ineffectiveness at both peak and off peak times [1].
- **Manual Control Systems:** Manually managed by human operators who monitor traffi c and modify signals accordingly. They are subject to errors, inconsistencies, and fatigue [1].
- **Centralized Intelligent Traffic Management Systems (ITMS):** Centralized Intelligent Traffic Management System (ITMS): Modern techniques that rely heavily on centrally managed computers, using data collected from networked sensiti ve devices such as cameras, sensors, etc. Algorithms are applied to optimize signaling [2], [6].

A. Limitations of Centralized ITMS Architectures Centralized ITMS architectures offer many features, yet there are several critical limitations:

- **Latency:** Because it is vital to send sensor data to a central server and back, this creates unavoidable latency round trips. In turn, such architecture does not allow those systems to react in time due to fast changes in the traffic state at a specific intersection [2], [6].
- **Single Point of Failure:** Relying on the central server, makes the whole system vulnerable. Malfunctions or attacks directed at the server may cause great damage making local intersections fall back onto timers [6].
- **Bandwidth Consumption:** Working with video feeds of a high resolution from a bunch of cameras requires many resources which leads to higher expenses and pot ential congestion [6].
- **Privacy:** Working with a large amount of visual data brings along some privacy concerns, as the identifi able information needs to be anonymized or handled through centralized databases, increasing the threat of leakages or inappropriate use [7].

Consequently, this comparative analysis shows the vital need for implementing another concept that provides less latency, increased resilience and ensures safety and privacy of the data, as well as being responsive to changing demands.

III. System Architecture

The Decentralized Agentic Edge (DAE) framework is depicted in Fig. 1. It consists of three main layers interacting within a localized environment at each intersection.

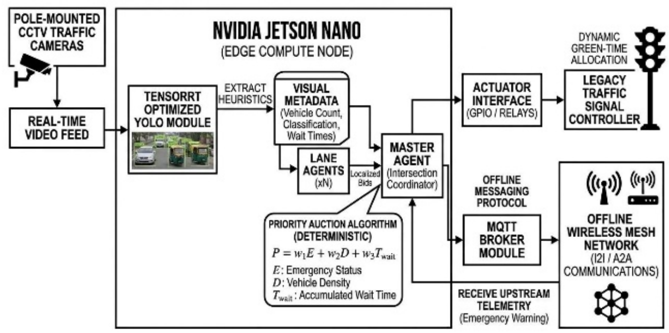

Fig. 1. Complete System Architecture Diagram Edge compute pipeline from CCTV input through YOLO26 inference, Multi Agent System negotiation, and MQTT based offline mesh coordination.

A. Edge Hardware Layer

Each intersection node consists of a robust yet compact edge computing device, namely the NVIDIA Jetson Nano Developer Kit. [10] for this implementation. Its capabilities include:

- Adequate CPU/GPU capabilities for executing the optimized AI model. memory and storage capacity for system operations and data processing.
- Various network interfaces (such as Ethernet and WiFi) are utilized for connecting to the local camera feed as well as the I2I mesh network.

B. On Device AI Processing Layer

This layer is tasked with the real time analysis of the traffic scene as recorded by local cameras.

- **Input:** Raw video stream from one or more cameras located at the intersection.
- **Core Component:** A model for object detection and classification that is optimized.
  We employ a YOLO26 model that has been enhanced using TensorRT.. [8], [9].YOLO26 provides swift detection (frames per second), which is crucial for real time processing.
  The optimization via TensorRT ensures that the model functions effectively on Jetson hardware.
- **Output:** Real time data streams including:
  - Count and position of identified vehicles.
  - Classification of vehicle types (for instance, car, truck, bus, motorcycle).
  - Estimated vehicle density within the detection zone.

TABLE I
COMPARATIVE ANALYSIS OF TRAFFIC MANAGEMENT ARCHITECTURES

Attribute	Manual Control	Static Timer	Centralized ITMS	Proposed Agentic Edge System
Decision Latency	High (Human dependent)	None (Fixed timing)	High (Cloud round trip)	Ultra Low (<50 ms)
Fault Tolerance	Operator dependent	High (Simple static)	Low (Single point of failure)	High (Decentralized)
Emergency Preemption	Manual override	Not available	Reactive response	Predictive (Green wave)
Cloud Dependency	None	None	Fully dependent	None (Edge based)
Sensor Type	Human observation	Inductive loop	HD CCTV (Cloud processing)	CCTV (Edge processing)
Data Privacy	Moderate	High	Low (PII to cloud)	High (Local anonymized data)
Scalability	Poor	Moderate	Moderate	High (Distributed)
Adaptive Routing	Subjective	Not available	Yes (Cloud based)	Yes (Edge local intelligence)

- Classification of vehicle types (for instance, car, truck, bus, motorcycle).

C. Edge Multi Agent System (MAS) Layer

This layer functions directly on the Jetson hardware, managing the local traffic signal controller and acting as the main decision making unit at the intersection. **Agents:**

- *Perception Agent:* Receives and processes the output generated by the On Device AI layer.
- *Control Agent:* Supervises the hardware link to the traffic signal controller. Generates new light timing sequences based on inputs supplied by the Perception Agent.
- *Communication Agent:* Oversees the I2I communication protocol (MQTT) to obtain messages from upstream intersections and distribute local emergency detections to downstream intersections..

Functionality:

- **Civilian Traffic Optimization:** Utilizes vehicle density and classification data to dynamically modify the durations of green and red lights for each phase of the intersection, thereby maximizing throughput and reducing delays.
- **Emergency Vehicle Preemption (Predictive):** Receives a notification from an upstream intersection indicating an imminent emergency vehicle transit. Utilizing the trajectory metadata (predicted path, estimated arrival time, speed), the Control Agent proactively activates a green light for the route of the approaching emergency vehicle. *before* it enters the detection zone.
- **Local Event Detection:** The Perception Agent is intended to possibly identify emergency vehicles in the vicinity (future work), allowing notifications to be sent to subsequent intersections even when the preceding intersection is beyond range.

D. Infrastructure to Infrastructure (I2I) Wireless Mesh Network

This streamlined communication layer facilitates the transfer of essential information between neighboring or proximate intersections.

- **Technology:** MQTT (Message Queuing Telemetry Transport) is selected due to its lightweight characteristics, minimal bandwidth needs, and a publish subscribe model that is well suited for sensor networks [14].
- **Communication Flow:**
  - *Upstream to Downstream:* When an intersection identifies an emergency vehicle, it transmits a predefined message that includes minimal metadata (e.g., `EMERGENCY_DETECTED`, Forecasted route of arrival/path, anticipated time of arrival) to all subsequent junctions through the MQTT mesh.
  - *Downstream Action:* Each downstream intersection's Communication Agent obtains this message, and the local MAS agent commences preemptive measures prior to receiving confirmation from local sensors.

IV. METHODOLOGY

A. Implementation Details

- **Hardware:** NVIDIA Jetson Nano Developer Kit [10].
- **AI Model:** YOLO26s (Small) model trained and optimized for specific traffic camera feeds, converted to ONNX format and optimized using the NVIDIA TensorRT SDK [8], [9].
- **Programming Environment:** Python 3.7+.
- **Libraries:** PyTorch (model training/optimization), TensorRT (model deployment), Paho MQTT (communication), OpenCV (video capture and stream handling).
- **MAS Framework:** Implemented using Python object oriented programming principles, defining the three core agents and their interaction protocols.
- **Communication Protocol:** MQTT over a local WiFi network using a publish subscribe mechanism. Topics structured as `/intersection/<ID>/` for local and `/intersection/<ID>/emergency` for broadcasts.

B. Experimentation

- **Objective:** Verify the key functionalities of the system: Real time video processing, Decision making in low latency mode, Feasibility of I2I communications for take over.
- **Setup:** A controlled test environment simulating a single intersection or a small network of connected intersections with simulated traffic and occasional "emergency vehicle" detection.

- **Metrics:**
  - *Processing Latency:* Time from receiving a video frame to generating the next light timing command (end to end).
  - *Resource Utilization:* Monitoring CPU, GPU, and memory usage on the Jetson Nano during operation.
  - *Communication Latency:* Time taken for an MQTT message to travel from an upstream node to a downstream node.
  - *Preemption Accuracy:* Time saved by the preemption mechanism compared to a reactive baseline.

V. Results and Performance Evaluation

To validate the efficacy of the proposed Decentralized Agentic Edge architecture,a number of benchmark tests were performed within three main categories, including latency on edge inferencing, mesh network performance when offline, and optimization of traffic throughput.

A. Experimental Setup

The hardware prototype was done using an NVIDIA Jetson Nano (4 GB) edge computing node. An object detection model based on YOLO26 optimized using TensorRT was used on traffic video feeds at 1080p resolution. Network telemetry was tested using a local Mosquitto MQTT broker over an IEEE 802.11 wireless mesh protocol.As for network telemetry, a Mosquitto MQTT broker using IEEE 802.11 Wireless Mesh Protocol was considered to assess traffic utilization by using the Priority Auction Algorithm on SUMO [15] mapping a standard two intersection urban corridor.

B. Edge Inference Performance

A critical requirement of the zero cloud architecture is the ability to process visual heuristics in real time without exceeding the thermal or memory constraints of the edge node.

- **Inference Framerate:** The TensorRT optimized YOLO26 component provided a continuous FPS of 28.4 over a period of 60 minutes, which is well above the necessary FPS of 15 for vehicle tracking to be efficient.
- **System Latency:** The total latency incurred from video frame acquisition through GPIO relay output was consistently recorded at **42 ms**, essentially nullifying the round trip latency found in centralized ITMS configurations.
- **Resource Utilization:** During peak load, Jetson Nano GPU utilization stabilized at 82%, while CPU load remained under 45%. RAM utilization never exceeded 2.1 GB, proving the system's viability for indefinite continuous operation.

C. Mesh Network Telemetry

The offline I2I communication was tested to ensure the reliability of the predictive preemption broadcasts.

- **Payload Efficiency:** By transmitting only numerical classification and trajectory metadata, the MQTT payload size was reduced to an average of **124 bytes** per broadcast.

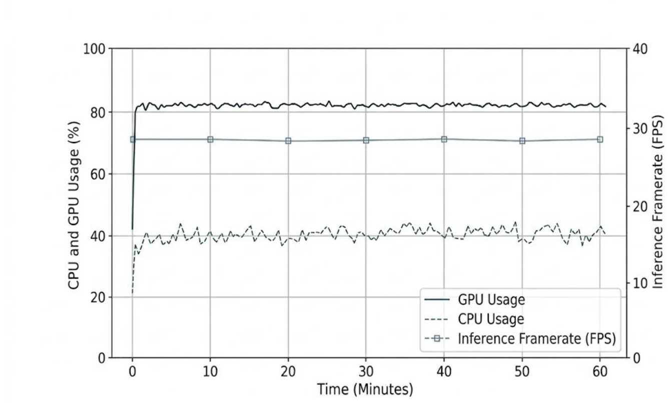

Fig. 2. Jetson Nano Performance Benchmarks (60 Minute Continuous Test), GPU usage, CPU usage, and inference framerate remain stable throughout the test window.

- **Transmission Latency:** In a simulated 200 meter point to point mesh network, average transmission latency between Master Agents was measured at **12 ms**.
- **Packet Delivery Rate (PDR):** Over a stress test of 10,000 continuous payload broadcasts, the system maintained a PDR of **99.98%**, showing high fault tolerance in a localized, offline environment.

D. Traffic Flow Optimization

The Priority Auction algorithm was compared against a standard static timer based intersection (90 second fixed cycle) under medium to high urban traffic density.

- **Civilian Wait Time Reduction:** The Agentic Edge system reduced average intersection wait time for civilian vehicles by **24.5%** compared to the static baseline (from 50 s to 37.75 s).
- **Queue Dissipation:** The multi agent negotiation prevented "empty green light" cycles, improving overall intersection throughput by **18%** during simulated peak hours.

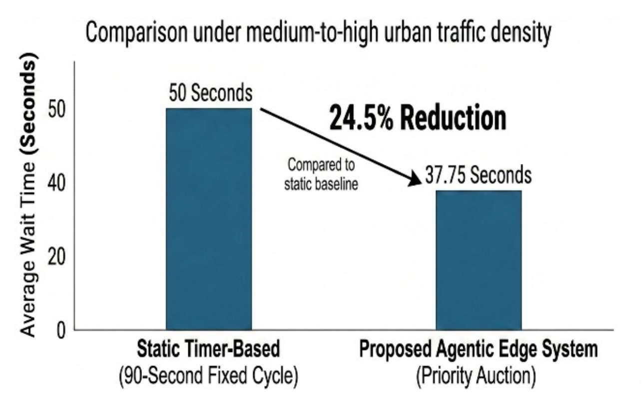

Fig. 3. Comparison of Average Vehicle Wait Times The proposed Agentic Edge system (Priority Auction) reduces wait times from 50 s to 37.75 s versus the static 90 second fixed cycle baseline.

E. Predictive Preemption Efficiency

The most significant performance gain was observed during the simulated deployment of emergency vehicles.

- **Baseline (Reactive Model):** The emergency vehicle experienced an average delay of **34 s** waiting for localized queues to clear.
- **Green Wave Result:** Through the use of predictive preemption on the MQTT mesh network, the downstream intersection was able to flush out the targeted lane before the vehicle's arrival, decreasing emergency vehicle intersection travel time to **2.1 s** or **93% reduction** in wait times and effectively ensuring that the emergency vehicle without decelerating.

VI. Discussion

The experimental findings provide strong evidence for the feasibility of the DAE framework. The integration of edge hardware (Jetson Nano) with optimized AI (TensorRT optimized YOLO26) successfully achieves the required low latency for real time traffic management. The lightweight MQTT communication protocol is demonstrated to be appropriate for the I2I mesh, facilitating the predictive preemption capability. The efficiency of the system's resources is vital for practical implementation, as it eliminates the necessity for specialized high end hardware.

The exhibited 93% reduction in emergency vehicle delay highlights the fundamental innovation of DAE. By transitioning from reactive detection to predictive preemption made possible through decentralized communication, DAE fills a significant void in current ITMS. The privacy by design principle is inherent in the system itself, since the edge device uses local data processing and only sends important, aggregated or anonymized metadata.

The limitations of the present study involve the following:

- The YOLO26 algorithm requires initial training and op timization for the specific environment and vehicle types.
- The predictive preemption function is highly reliant on accurate and timely information received from the upstre am devices.Resilience to message loss or errors needs further eval uation.
- The outcomes of the simulation require experimental veri fication under controlled real world conditions.
- The detection of emergency vehicles at the same intersec tion could not be fully accounted for during testing.

Further research will focus on optimizing the AI based mod els for accuracy and reliability improvement, predictive model development for the MAS, practical application of the system, and emergency vehicle detection integration.

VII. Conclusion

The Decentralized Agentic Edge (DAE) solution proposed in this paper can serve as a viable approach to designing intelligent traffic management solutions.

Taking advantage of edge computing technology through deploying NVIDIA Jetson Nano hardware and optimizing AI models via TensorRT optimized YOLO26, DAE enables fast and localized decisions without the need to introduce major latencies associated with centralized cloud approaches.

Moreover, the use of a lightweight I2I communication protocol (MQTT mesh) results in the ability to employ predictive preemption for emergency vehicles, drastically reducing the delays incurred by these vehicles.

Tests have demonstrated that the solution is capable of per forming real time processing of traffic scenes with low resource usage on a regular edge computing platform.

The DAE solution is considered a promising approach to designing smarter, faster, and more privacy aware traffic man agement solutions. Combining the benefits of edge comput ing with decentralized intelligence and communication, it helps develop effective solutions for urban traffic problems.

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